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Power transmission unit and brake systems using same

Abstract

A power transmission unit of a single-acting plunger type includes a ball screw, a ball nut driven by the ball screw, and a plunger piston coupled to the ball nut. A block housing at least partially encloses the plunger piston and the ball screw. A plunger chamber is at least partially defined by the block housing and a face of the plunger piston. The plunger chamber is selectively pressurized by reciprocal motion of the plunger piston with respect to the block housing driven by longitudinal motion of the ball nut. An electric motor selectively drives the ball screw to responsively reciprocate the plunger piston within the plunger chamber. A motor housing at least partially encloses the ball nut, the electric motor, and the plunger piston. The motor housing includes a block lip which is crimped to the block housing to maintain the motor housing in relation to the block housing.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a power transmission unit and, more particularly, to a brake system for hydraulically actuating two pairs of brakes using the power transmission unit in a hydraulic brake by wire system.

BACKGROUND

(2) A brake system may include a plurality of wheel brakes and a hydraulic braking pressure generator, a braking pressure modulator which is provided in the pressure fluid conduits between the braking pressure generator and the wheel brakes and which serves to vary the braking pressure by changing the volume of a chamber containing the hydraulic fluid, sensors for determining the wheel rotational behavior, and electronic circuits for processing the sensor signals and for generating braking-pressure control signals. Brake systems may also include an electronic control unit that can be used to provide a braking command to the wheel brakes, autonomously and/or manually (e.g., via the use of an operator-manipulable brake pedal).

SUMMARY

(3) In an aspect, a power transmission unit of a single-acting plunger type is disclosed. The power transmission unit includes a ball screw, a ball nut selectively driven by the ball screw for longitudinal motion relative thereto, and a plunger piston operatively coupled to the ball nut. A block housing at least partially encloses the plunger piston and the ball screw. A plunger chamber is at least partially defined by the block housing and a face of the plunger piston. The plunger chamber is configured to contain hydraulic fluid and is selectively pressurized by reciprocal motion of the plunger piston with respect to the block housing driven by longitudinal motion of the ball nut. An output channel selectively places the plunger chamber in fluid communication with at least one component of a brake system including the power transmission unit. An electric motor selectively drives the ball screw to responsively reciprocate the plunger piston within the plunger chamber. A motor housing at least partially encloses the ball nut, the electric motor, and the plunger piston. The motor housing includes a block lip which is crimped to the block housing to maintain the motor housing in relation to the block housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding, reference may be made to the accompanying drawings, in which:

(2) FIG. 1 is a schematic cross-sectional view of a component of a brake system according to an aspect of the present invention, in a first example configuration;

(3) FIG. 2 is a schematic cross-sectional view of the component of FIG. 1, in a second example configuration;

(4) FIG. 3 is a schematic partial view of an example hydraulic brake by wire unit including the component of FIGS. 1-2;

(5) FIG. 4 is a schematic front perspective view of the hydraulic brake by wire unit of FIG. 3;

(6) FIG. 5 is a schematic rear perspective view of the hydraulic brake by wire unit of FIG. 3;

(7) FIG. 6 is a schematic left side view of the hydraulic brake by wire unit of FIG. 3;

(8) FIG. 7 is a schematic right side view of the hydraulic brake by wire unit of FIG. 3;

(9) FIG. 8 is a schematic front view of the hydraulic brake by wire unit of FIG. 3;

(10) FIG. 9 is a schematic partial bottom view of the hydraulic brake by wire unit FIG. 3;

(11) FIG. 10 is a schematic front perspective view of the hydraulic brake by wire unit of FIG. 3 in a first alternate configuration;

(12) FIG. 11 is a schematic rear perspective view of the hydraulic brake by wire unit of FIG. 3 in the first alternate configuration;

(13) FIG. 12 is a schematic left side view of the hydraulic brake by wire unit of FIG. 3 in the first

alternate configuration;

(14) FIG. 13 is a schematic right side view of the hydraulic brake by wire unit of FIG. 3 in the first alternate configuration;

(15) FIG. 14 is a schematic front view of the hydraulic brake by wire unit of FIG. 3 in a second alternate configuration;

(16) FIG. 15 is a schematic partial bottom view of the hydraulic brake by wire unit of FIG. 3 in the second alternate configuration; and

(17) FIG. 16 is a schematic diagram of a brake system including the component of FIGS. 1-2.

DESCRIPTION OF ASPECTS OF THE DISCLOSURE

(18) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of ordinary skill in the art to which the present disclosure pertains.

(19) The invention comprises, consists of, or consists essentially of the following features, in any combination.

(20) FIGS. 1-2 depict a power transmission unit **100** of a single-acting plunger type. The power transmission unit **100** includes a ball screw **102** and a ball nut **104** selectively driven by the ball screw **102** for longitudinal motion relative thereto. The “longitudinal” direction, as referenced herein, is shown by arrow “L” and is substantially horizontal, in the orientation of FIGS. 1-2.

(21) A plunger piston **106** is operatively coupled to the ball nut **104** for relative longitudinal movement therewith responsive to rotational movement, about the longitudinal direction, by the ball screw **102**. A block housing **108** at least partially encloses the plunger piston **106** and the ball screw **102**. A plunger chamber **110** is at least partially defined by the block housing **108** and a face **112** of the plunger piston **106**. The plunger chamber **110** is configured to contain hydraulic fluid and is selectively pressurized by reciprocal motion of the plunger piston **106** with respect to the block housing **108**, driven by longitudinal motion of the ball nut **104**. It is noted that the plunger piston **106** is shown in FIGS. 1-2 as being in a “home” or fully-retracted “rest” position, in which the plunger piston **106** is not exerting pressure on fluid within the plunger chamber. It is contemplated that an end cap **114** could be provided as a “liner” or otherwise to assist with maintaining a substantially fluidtight condition of the plunger chamber **110**, as desired.

(22) An output channel **116** selectively places the plunger chamber **110** in fluid communication with at least one component of a brake system including the power transmission unit **100**.

Pressurized hydraulic fluid is generated within the plunger chamber **110** and flows outward through the output channel **116** for providing actuating/motive pressure to components of any brake system, as desired, which includes the power transmission unit **100**. One of ordinary skill in the art will be readily able to configure a hydraulic brake system to use the power transmission unit **100** disclosed herein, particularly when a single-acting plunger type power transmission unit is desired for that specific hydraulic brake system.

(23) An electric motor **118** is provided for selectively driving the ball screw **102** to responsively reciprocate the plunger piston **106** within the plunger chamber **110**. A motor housing **120** at least partially encloses the ball nut **104**, the electric motor **118**, and/or the plunger piston **106**. The electric motor **118** includes a motor rotor **122** operatively coupled to the ball screw **102** for rotation therewith, and a motor stator **124** mounted on the motor housing **120** and stationary relative thereto. Torque developed by the motor stator **124** is resisted and/or “dissipated” by at least a portion of the motor housing **120**.

(24) A bearing **126** is provided for rotatably supporting a supported end **128** of the ball screw **102**, the supported end **128** being longitudinally spaced apart from the plunger chamber **110**. The motor housing **120** includes a bearing recess **130** with a bearing lip **132** crimped onto at least a portion of the bearing **126** to maintain the bearing **126** relatively stationary in relation to the motor housing **120**, as well as to selectively transmit or “dissipate” force from the bearing **126** into the motor housing **120**. The crimping of the bearing lip **132** onto the motor housing can be accomplished via

a plurality of discrete crimp locations spaced about a periphery of the bearing lip **132**, or can be done about substantially an entire circumferential extent of the bearing lip **132**. Particularly in the latter case, forces between the bearing **126** and the motor housing **120** can be transmitted efficiently and without unevenly spread local stress concentrations as might be developed in the prior art bolted arrangements for similar components.

(25) The motor housing **120** may include a block lip **134** which is crimped to the block housing **108** to maintain the motor housing **120** in relation to the block housing **108**. Because the power transmission unit **100** only builds pressure in one direction (i.e., is a “single-acting” plunger type unit), a “can” type motor housing **120** may be used, and the pressure-application direction will tend to reinforce or pull on the block lip **134** crimping arrangement, thus beneficially further seating the joint. If dual-direction pressure were to be present in the power transmission unit **100**, the block lip **134** crimping arrangement could undergo undesirable reverse-direction force which might tend to degrade the joint between the block lip **134** and the block housing **108** and/or provide a high-stress concentration adjacent the motor bearing area of the housing.

(26) For similar reasons to those discussed above with respect to the bearing lip **132**, use of the block lip **134** instead of a plurality of bolts, spot-welds, or other point-contact fasteners may assist with distributing force evenly about the circumference of the motor housing **120**. Additionally, for both the block lip **134** and bearing lip **132** crimped arrangements, use of the crimped-circumference scheme (or crimped-perimeter, when the components are not circular or curvilinear) can help with space arrangement considerations inside the power transmission unit and with respect to neighbouring structures, by eliminating the need for space for the bolts (or other point-contact fasteners) and space to allow for bolt manipulation. It is contemplated that welding, adhesives, fasteners, or any other suitable connection scheme or combination thereof could be used instead of, or in addition to, crimping, when desired.

(27) The plunger chamber **110** (or the end cap **114**, when present) may define a longitudinally extending central chamber axis (“CCA”, in the Figures) which is laterally offset (i.e., non-coaxial) from a longitudinally extending central plunger axis (“CPA”, in the Figures) of the plunger piston **106**. The “lateral” direction is substantially perpendicular to the longitudinal direction, and is substantially vertical, in the orientation of FIGS. 1-2). Stated differently, the plunger chamber **110** and/or the end cap **114** is “off-center” from the plunger piston **106**, as can be seen in FIG. 2. This offset placement may assist with providing desired flow of pressurized hydraulic fluid out of the power transmission unit **100**, since the annular space around the extended piston **106** in the plunger chamber **110** will be wider/thicker on the side of the plunger chamber **110** which is adjacent the output channel **116**. Such offset placement may also minimize the outer diameter of the end cap **114** to help reduce overall package size, in some use environments.

(28) A vent **136**, optionally including a “gore membrane”, may be provided in a known manner to assist with letting air or other gases vent from the plunger chamber **110**. When present, the vent **136** can help keep at least liquid water out of the inside volume of the motor housing **120** (water vapor may be permitted to pass the vent **136** in some arrangements) and can also keep pressure near-atmospheric in the area of the power transmission unit **100** adjacent the electric motor **118**.

(29) The plunger piston **106** may define a longitudinal internal plunger cavity **138** for receiving at least a portion of the ball screw **102** therein, thus facilitating a smaller overall longitudinal dimension of the power transmission unit. The internal plunger cavity **138** may include a force dissipator **140** at a blind end thereof. The force dissipator **140** may be configured for selective contact with a cantilevered or free end **142** of the ball screw **102**. When present, the force dissipator **140** may assist with absorbing and/or dissipating a longitudinally-directed force resulting from contact between the cantilevered end **142** of the ball screw with the inside (and especially the blind end) of the internal plunger cavity **138**. For example, if there is a loss of power to the electric motor **118** or other relevant components of the brake system, the ball screw **102** may “free spin” in the ball nut **104** and return the attached plunger piston **106**, potentially quite forcefully, to the “home”

position shown in FIGS. 1-2 under influence of pressurized hydraulic fluid within the plunger chamber **110**. (The plunger piston **106** could also or instead return to the “home”, or fully retracted, position upon startup of the brake system, as desired.) Depending upon the tolerances and relative configurations of the components of the power transmission unit **100**, the cantilevered end **142** of the ball screw **102** could “bang” longitudinally against the inside of the internal plunger cavity **138** (though a space therebetween is shown in FIG. 1), potentially causing unwanted forces and noise to develop within the power transmission unit **100**. The force dissipator **140**, when present, resists these unwanted effects of contact between the ball screw **102** and the plunger piston **106**.

(30) As shown schematically in FIGS. 1-2, the cantilevered end **142** of the ball screw **102** may include a slightly convex profile, configured to concentrate contact with the force dissipator **140** at a predetermined area of the convex profile (e.g., the area at/adjacent the marking asterisk, in FIGS. 1-2). Through such intentional force concentration, the force can be directed to, e.g., an intentionally hardened portion of the cantilevered end **142** of the ball screw **102**. Accordingly, this near-point contact between the intentional force concentration may be helpful in providing less frictional torque for a predetermined axial force between the contacting components. If frictional torque is more than the motor **118** can overcome, then the ball screw **102** will be prevented from turning, thus “locking up” the power transmission unit **100** in an undesirable manner.

(31) With specific reference now to FIG. 2, a configuration of the power transmission unit **100** is shown in which the force dissipator **140** includes a spring member **144**. The spring member **144** could be a simple coil spring, Belleville washer, and/or any other suitable spring member, and can readily be provided for a particular use environment of the present invention by one of ordinary skill in the art. However, as shown in FIG. 2, the spring member **144** of the force dissipator **140** includes a resilient bumper **146**. For example, a rubber puck or other suitable structure could be used as a resilient bumper **146**.

(32) The spring member **144** of the force dissipator **140** of FIG. 2 also may include a planar member **148** interposed longitudinally between the resilient bumper **146** and the cantilevered end **142** of the ball screw **102**. The planar member **148** may be, for example, a steel plate in some use environments of the power transmission unit **100**, and may assist both with protecting the resilient bumper **146** from erosion by repeated contact against the cantilevered end **142** of the ball screw **102** and with spreading force from the cantilevered end **142** about a wider area of the force dissipator **140** than just the portion thereof which contacts the predetermined area (asterisked in FIGS. 1-2). FIGS. 3-16 schematically depict the power transmission unit **100** of FIG. 1, for the sake of example, but one of ordinary skill in the art will understand that the power transmission unit **100** configured as shown in FIG. 2 could also or instead be used as desired.

(33) Turning now to FIG. 3, a hydraulic brake by wire (“HBW”) unit **152** is schematically depicted. The HBW unit **152** of FIG. 3 includes a power transmission unit **100** (which, again, could be configured as in either of FIGS. 1-2 or in any other desired manner). First and second multiplex control valve arrangements **154A**, **154B** are fluidly connected to the output channel **116** of the power transmission unit **100**. Each of the first and second multiplex control valve arrangements **154A**, **154B** is interposed hydraulically between the power transmission unit **100** and a respective one of first and second iso tube ports **156A** and **156B**. The first and second iso tube ports **156A** and **156B** are configured for fluidic attachment to respective first and second wheel brakes (labeled here, by way of example, as left front and right front hydraulically actuated wheel brakes). The first and second iso tube ports **156A** and **156B** may be threaded, clipped, and/or swaged onto the channels connecting them with other components of the HBW unit **152**, or formed in any other desired fashion. The first and second multiplex control valve arrangements **154A**, **154B** each include respective first and second iso-type valves **158** and **160**. The HBW unit **152** could be configured for use in any suitable brake system, with an example brake system being discussed below and with reference to FIG. 16. Other nonlimiting examples, among myriad systems, of brake systems in which the HBW unit **152** could be suitably used (with the power transmission unit **100**

described herein substituted for the power transmission unit shown in the referenced applications) are disclosed in U.S. patent application Ser. No. 17/366,715, filed 2 Jul. 2021 and titled “Apparatus and Method for Selectively Actuating Wheel Brakes of a Hydraulic Brake System”; U.S. patent application Ser. No. 17/400,210, filed concurrently herewith and titled “Fluid Separator and Brake Systems Using Same”, hereafter referenced as “the Ser. No. 17/400,210 application”; and U.S. patent application Ser. No. 17/400,250, filed concurrently herewith and titled “Apparatus and Method for Control of a Hydraulic Brake System including Manual Pushthrough” all of which are incorporated herein by reference in their entirety.

(34) A reservoir port **162** is fluidly connected to the output channel **116** of the power transmission unit **100** and is configured to selectively place the output channel **116** (and other components of the HBW unit **152**) into fluid connection with a reservoir of the brake system. A venting valve **164** and a pressure sensor **166** may be fluidly connected to the output channel **116** of the power transmission unit **100**.

(35) Any desired type of position sensor (shown schematically at **168**) is configured to produce a position signal indicative of a location of the plunger piston **106** relative to the plunger chamber **110** may be provided, such as, but not limited to, a Hall Effect sensor, an inductive sensor, a motor rotation counter, a time-of-flight sensor, any other desired sensor type, or any combination thereof. Regardless of the nature of the sensing scheme, the position sensor **168**, when present, generates a position signal responsive to a sensed and/or calculated location of the plunger piston **106** with respect to the plunger chamber **110**. The position sensor **168** may also be used to provide electronic commutation of the motor **118**, especially when a brushless DC motor is used. An electronic control unit receives the position signal from the position sensor **168** and responsively controls fluid flow between the plunger chamber **110** and at least one of the first iso tube port **156A**, the second iso tube port **156B**, and the reservoir port **162**. This control by the electronic control unit may be accomplished via manipulation of one or more of the iso valves **158**, the iso valves **160**, and the venting valve(s) **164** of the HBW unit **152**.

(36) FIG. 3 represents a schematic “diagram view” of the HBW unit **152**. However, as one of ordinary skill in the art will be aware, components and fluid channels/lumens of an HBW unit **152** and/or a brake system may be collectively machined, molded, inserted, formed in, or otherwise provided to a common block housing **108**, which often becomes quite internally complex and labyrinthine in order to accommodate the position, number, and size of multiple components in a compact manner. FIGS. 4-15 depict various configurations of HBW units **152** in a form which might be crafted for insertion into a vehicle (though these Figures, like all Figures of the present application, are not drawn to scale). Due to the difficulty in showing internal passages of a solid block, FIGS. 4-15 show the various ports, valves, channels, and other components of the HBW unit **152** as being “freestanding” structures, floating in space lightly and schematically delineated as being the block housing **108**. One of ordinary skill in the art will realize, however, that the depicted “positive space” structures are often, in reality, likely going to be “negative space” voids within a block of material, which is omitted here for clarity.

(37) FIGS. 4-9 depict various views of the HBW unit **152**, showing that the first and second multiplex control valve arrangements **154A**, **154B**, the first and second iso tube ports **156A**, **156B**, the venting valve **164**, and/or the pressure sensor **156** are collectively contained at least partially within the block housing **108** which also contains the power transmission unit **100**. (Again, the block housing **108** is largely absent from FIGS. 4-9 for clarity, but one of ordinary skill in the art will understand the manner in which the block housing **108** encloses the depicted components).

(38) As previously mentioned, the HBW unit **152** or components thereof may be in signal exchanging communication with an electronic control unit **170**. The electronic control unit (“ECU”) **170** may be configured for attachment to the block housing **108** at a location longitudinally opposite the block housing **108** from the motor housing **120**, as shown in FIGS. 4-5. A connector **172** may be provided for routing signals and/or power to and from the ECU **170**. The

block housing may include at least one signal passageway (shown schematically at **174**) configured for routing a signal wire between the ECU **170** and the electric motor **118** for passing at least one of electric power and control signals therebetween. It should be noted here that relative placement of the ECU **170**, motor housing **120**, connector **172**, signal passageway **174**, and other components of the HBW unit **152** may be of particular interest for certain use environments. For example, and as shown in it least FIGS. **4-9**, certain components of the HBW unit **152** are spaced apart from the bottom of the unit (in the intended use orientation), to avoid locating them at/near a leakage pocket which could be present in the bottom of the unit. Additionally, placement of at least one signal passageway **174** adjacent the ECU connector **172** may be helpful, in some configurations of the HBW unit **152**, for both power routing and power component placement on a circuit board associated with the ECU **170**.

(39) FIGS. **10-15** schematically depict an HBW unit **152** including an attached ECU **170**, as well as an attached diagonal housing **176**. As disclosed in the Ser. No. 17/400,210 application, a three-wheel failure mode brake system can be provided wherein a single power transmission unit **100** can hydraulically actuate two wheel brakes (e.g., via first and second iso tube ports **156A**, **156B** as mentioned above) during normal non-failure braking mode, as well as a third wheel brake in a backup braking mode via a fluid separator **178**. Accordingly, the HBW unit **152** of FIGS. **10-15** could be used in a modular manner to provide one of the two pressure circuits in, for example, the embodiment depicted in FIG. 2 of the Ser. No. 17/400,210 application.

(40) As shown in FIGS. **10-15**, at least one jumper tube **180** could be used to attach the diagonal housing **176** to the HBW unit **152** when desired. (Certain components of the HBW unit **152** are left unlabeled in FIGS. **10-15** for clarity, and can be identified with reference to FIGS. **4-9**.) An additional multiplex control valve arrangement (shown generally at **154C** and discussed below) helps to control fluid flow through the fluid separator **178** to the “third” wheel brake via iso tube port **156C**. As shown here, the HBW unit **152** directly controls the right front wheel brake via iso tube port **156A** and the left rear wheel brake via iso tube port **156B**, in the non-failure normal braking mode, and controls the left front wheel brake via iso tube port **156C** in the backup braking mode. The fluid separator **178** also receives fluid from a second HBW unit via SAP port **182**, as discussed at length in the Ser. No. 17/400,210 application.

(41) FIG. **16** schematically depicts a brake system **184** for selectively actuating two pairs of hydraulically actuated wheel brakes **186** of a vehicle. The system includes a reservoir **188** and a first HBW unit **152A**, which includes a first power transmission unit **100A**. The reservoir port **162** of the first HBW unit **152A** is in fluid communication with the reservoir **188**, and the first and second iso tube ports **156A**, **156B** of the first HBW unit **152A** are in fluid communication with respective wheel brakes **186** of a selected pair of hydraulically actuated wheel brakes **186A**, **186D** of the vehicle. The first HBW unit **152A** is configured to selectively generate pressurized hydraulic fluid for actuating the selected pair of the hydraulically actuated wheel brakes **186A**, **186D** during a normal non-failure braking mode.

(42) The brake system **184** also includes a second HBW unit **152B**, which includes a second power transmission unit **100B**. The reservoir port **162** of the second HBW unit **152B** is in fluid communication with the reservoir **188**, and the first and second iso tube ports **156A**, **156B** of the second HBW unit **152B** are in fluid communication with respective wheel brakes **186** of an other pair of hydraulically actuated wheel brakes **186B**, **186C** of the vehicle (different than the selected pair). The second HBW unit **152B** is configured to selectively generate pressurized hydraulic fluid for actuating the other pair of the hydraulically actuated wheel brakes **186B**, **186C** during a normal non-failure braking mode. At least one electronic control unit **170** of the brake system **184** (two shown, for redundancy) is configured to control at least one of the first and second power transmission units **100**.

(43) The brake system **184** also includes a deceleration signal transmitter **190** including a brake pedal **192** connected to a brake pedal unit **194** and selectively actuated by a driver of the vehicle to

indicate a desired braking command. The brake pedal unit **194** has a brake sensor **196** for determining a position of the brake pedal **192** and responsively producing a braking signal corresponding to the desired braking command. The braking signal is transmitted to the at least one electronic control unit **170**. The at least one electronic control unit **170** controls at least one of the first and second power transmission units **100A**, **100B** responsive to the braking signal.

(44) Though omitted from FIG. **16**, it is contemplated that at least a selected one of the first and second HBW units **152A**, **152B** may include a diagonal housing **176** like that shown in FIGS. **10-15**. As mentioned with respect to those Figures, and referencing the Ser. No. 17/400,210 application, the diagonal housing **176** may include a fluid separator **178** and a third multiplex control valve arrangement **154C**, which could include third and fourth iso-type valves in some use environments, but here is shown as being a single multiplex valve **154C** providing a multiplex function when the brake system **184** is in a backup braking mode. The diagonal housing **176** may also include an iso tube port **156C** and an SAP port **182** for connection to the power transmission unit **100** of the other one of the first and second HBW units **152A**, **152B**. The diagonal housing **176** may be configured for selective attachment to the block housing **108** of the selected one of the first and second HBW units **152A**, **152B** at a location spaced from the motor housing **120**. The diagonal housing **176** may also be configured for selective fluid attachment to at least one connector tube (e.g., jumper tube **180**) of the block housing **108** for fluid communication of at least the fluid separator **178**, the third multiplex control valve arrangement **154C**, and the iso tube port **156C** with the output channel **116** of the selected one of the first and second HBW units **152A**, **152B**.

(45) When present, the diagonal housing **176** may be associated with the selected one of the first and second HBW units **152A**, **152B** for selective control, in a backup braking mode, of at least one wheel brake **186** of the pair of wheel brakes which is opposite the wheel brake **185** in the pair of wheel brakes which is otherwise associated with the selected one of the first and second HBW units **152A**, **152B** for operation in a normal non-failure braking mode. That is, the backup braking control of at least one chosen wheel brake **186** is controlled/actuated by the opposite HBW unit **152** from that which controls/actuates the chosen wheel brake **186** for normal non-failure braking, similar to the scheme disclosed in the Ser. No. 17/400,210 application.

(46) As used herein, the singular forms “a”, “an”, and “the” can include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising”, as used herein, can specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

(47) As used herein, the term “and/or” can include any and all combinations of one or more of the associated listed items.

(48) It will be understood that when an element is referred to as being “on”, “attached” to, “connected” to, “coupled” with, “contacting”, “adjacent”, etc., another element, it can be directly on, attached to, connected to, coupled with, contacting, or adjacent the other element, or intervening elements may also be present. In contrast, when an element is referred to as being, for example, “directly on”, “directly attached” to, “directly connected” to, “directly coupled” with, “directly contacting”, or “directly adjacent” another element, there are no intervening elements present. It will also be appreciated by those of ordinary skill in the art that references to a structure or feature that is disposed “directly adjacent” another feature may have portions that overlap or underlie the adjacent feature, whereas a structure or feature that is disposed “adjacent” another feature might not have portions that overlap or underlie the adjacent feature.

(49) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “proximal”, “distal”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms can encompass different orientations of a device in use or operation, in addition to the orientation depicted in the figures. For example, if a device in the

figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features.

(50) As used herein, the phrase “at least one of X and Y” can be interpreted to include X, Y, or a combination of X and Y. For example, if an element is described as having at least one of X and Y, the element may, at a particular time, include X, Y, or a combination of X and Y, the selection of which could vary from time to time. In contrast, the phrase “at least one of X” can be interpreted to include one or more Xs.

(51) It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a “first” element discussed below could also be termed a “second” element without departing from the teachings of the present disclosure. The sequence of operations (or steps) is not limited to the order presented in the claims or figures unless specifically indicated otherwise.

(52) While aspects of this disclosure have been particularly shown and described with reference to the example aspects above, it will be understood by those of ordinary skill in the art that various additional aspects may be contemplated. For example, the specific methods described above for using the apparatus are merely illustrative; one of ordinary skill in the art could readily determine any number of tools, sequences of steps, or other means/options for placing the above-described apparatus, or components thereof, into positions substantively similar to those shown and described herein. In an effort to maintain clarity in the Figures, certain ones of duplicative components shown have not been specifically numbered, but one of ordinary skill in the art will realize, based upon the components that were numbered, the element numbers which should be associated with the unnumbered components; no differentiation between similar components is intended or implied solely by the presence or absence of an element number in the Figures. Any of the described structures and components could be integrally formed as a single unitary or monolithic piece or made up of separate sub-components, with either of these formations involving any suitable stock or bespoke components and/or any suitable material or combinations of materials. Any of the described structures and components could be disposable or reusable as desired for a particular use environment. Any component could be provided with a user-perceptible marking to indicate a material, configuration, at least one dimension, or the like pertaining to that component, the user-perceptible marking potentially aiding a user in selecting one component from an array of similar components for a particular use environment. A “predetermined” status may be determined at any time before the structures being manipulated actually reach that status, the “predetermination” being made as late as immediately before the structure achieves the predetermined status. The term “substantially” is used herein to indicate a quality that is largely, but not necessarily wholly, that which is specified—a “substantial” quality admits of the potential for some relatively minor inclusion of a non-quality item. Though certain components described herein are shown as having specific geometric shapes, all structures of this disclosure may have any suitable shapes, sizes, configurations, relative relationships, cross-sectional areas, or any other physical characteristics as desirable for a particular application. Any structures or features described with reference to one aspect or configuration could be provided, singly or in combination with other structures or features, to any other aspect or configuration, as it would be impractical to describe each of the aspects and configurations discussed herein as having all of the options discussed with respect to all of the other aspects and configurations. A device or method incorporating any of these features should be understood to fall under the scope of this disclosure as determined based upon the claims below and any equivalents thereof.

(53) Other aspects, objects, and advantages can be obtained from a study of the drawings, the disclosure, and the appended claims.

Claims

1. A power transmission unit of a single-acting plunger type, the power transmission unit comprising: a ball screw; a ball nut selectively driven by the ball screw for longitudinal motion relative thereto; a plunger piston operatively coupled to the ball nut; a block housing at least partially enclosing the plunger piston and the ball screw; a plunger chamber at least partially defined by the block housing and a face of the plunger piston, the plunger chamber being configured to contain hydraulic fluid and selectively pressurized by reciprocal motion of the plunger piston with respect to the block housing driven by longitudinal motion of the ball nut; an output channel selectively placing the plunger chamber in fluid communication with at least one component of a brake system including the power transmission unit; an electric motor for selectively driving the ball screw to responsively reciprocate the plunger piston within the plunger chamber; a motor housing at least partially enclosing the ball nut, the electric motor, and the plunger piston, the motor housing including a block lip which is crimped to the block housing to maintain the motor housing in relation to the block housing; and a bearing for rotatably supporting a supported end of the ball screw spaced apart from the plunger chamber, wherein the motor housing includes a bearing recess with a bearing lip crimped onto at least a portion of the bearing to maintain the bearing in relation to the motor housing.
2. The power transmission unit of claim 1, wherein the plunger chamber defines a longitudinally extending central chamber axis which is laterally offset from a longitudinally extending central plunger axis of the plunger piston.
3. The power transmission unit of claim 1, wherein the electric motor includes a motor rotor operatively coupled to the ball screw and a motor stator mounted on the motor housing wherein torque developed by the motor stator is resisted by at least a portion of the motor housing.
4. A hydraulic brake by wire unit, comprising: the power transmission unit of claim 1; first and second multiplex control valve arrangements fluidly connected to the output channel of the power transmission unit, each of the first and second multiplex control valve arrangements being interposed hydraulically between the power transmission unit and a respective one of first and second iso tube ports, the first and second multiplex control valve arrangements each including respective first and second iso-type valves.
5. The hydraulic brake by wire unit of claim 4, including a reservoir port fluidly connected to the output channel of the power transmission unit.
6. The hydraulic brake by wire unit of claim 4, wherein the first and second multiplex control valve arrangements and the first and second iso tube ports are collectively contained at least partially within the block housing of the power transmission unit.
7. The hydraulic brake by wire unit of claim 4, including a venting valve and a pressure sensor fluidly connected to the output channel of the power transmission unit.
8. The hydraulic brake by wire unit of claim 7, wherein the first and second multiplex control valve arrangements, the first and second iso tube ports, the venting valve, and the pressure sensor are collectively contained at least partially within the block housing of the power transmission unit.
9. The hydraulic brake by wire unit of claim 4, including an electronic control unit configured for attachment to the block housing at a location longitudinally opposite the block housing from the motor housing, wherein the block housing includes at least one signal passageway configured for routing a signal wire between the electronic control unit and the electric motor for passing at least one of electric power and control signals therebetween.
10. The hydraulic brake by wire unit of claim 4, including a position sensor configured to produce a position signal indicative of a location of the plunger piston relative to the plunger chamber.
11. The hydraulic brake by wire unit of claim 10, wherein the position sensor generates a position signal responsive to a sensed location of the plunger piston with respect to the plunger chamber,

and the electronic control unit receives the position signal from the position sensor and responsively controls fluid flow between the plunger chamber and at least one of the first iso tube port, the second iso tube port, and the reservoir port.

12. A brake system for selectively actuating two pairs of hydraulically actuated wheel brakes of a vehicle, the system comprising: a reservoir; a first hydraulic brake by wire unit of claim 4, with the reservoir port of the first hydraulic brake by wire unit being in fluid communication with the reservoir and the first and second iso tube ports of the first hydraulic brake by wire unit being in fluid communication with respective wheel brakes of a selected pair of hydraulically actuated wheel brakes of the vehicle, the first hydraulic brake by wire unit being configured to selectively generate pressurized hydraulic fluid for actuating the selected pair of the hydraulically actuated wheel brakes during a normal non-failure braking mode; a second hydraulic brake by wire unit of claim 10, with the reservoir port of the second hydraulic brake by wire unit being in fluid communication with the reservoir and the first and second iso tube ports of the second hydraulic brake by wire unit being in fluid communication with respective wheel brakes of an other pair of hydraulically actuated wheel brakes of the vehicle, the second hydraulic brake by wire unit being configured to selectively generate pressurized hydraulic fluid for actuating the other pair of the hydraulically actuated wheel brakes during a normal non-failure braking mode; and at least one electronic control unit configured to control at least one of the first and second power transmission units.

13. The brake system of claim 12, including a deceleration signal transmitter including a brake pedal connected to a brake pedal unit and selectively actuated by a driver of the vehicle to indicate a desired braking command, the brake pedal unit having a brake sensor for determining a position of the brake pedal and responsively producing a braking signal corresponding to the desired braking command, the braking signal being transmitted to the at least one electronic control unit, and the at least one electronic control unit controlling at least one of the first and second power transmission units responsive to the braking signal.

14. The brake system of claim 12, wherein at least a selected one of the first and second hydraulic brake by wire units includes a diagonal housing, the diagonal housing including a fluid separator, a third multiplex control valve arrangement including at least one multiplex valve, an iso tube port; and an SAP port for connection to the power transmission unit of the other one of the first and second hydraulic brake by wire units, the diagonal housing being configured for selective attachment to the block housing at a location spaced from the motor housing and selective fluid attachment to a connector tube of the block housing for fluid communication of at least the fluid separator, the third multiplex control valve arrangement, and the iso tube port with the output channel; wherein the diagonal housing is associated with the selected one of the first and second hydraulic brake by wire units for selective control, in a backup braking mode, of at least one wheel brake of the pair of wheel brakes which is opposite the wheel brake in the pair of wheel brakes which is otherwise associated with the selected one of the first and second hydraulic brake by wire units for operation in a normal non-failure braking mode.

15. A power transmission unit of a single-acting plunger type, the power transmission unit comprising: a ball screw; a ball nut selectively driven by the ball screw for longitudinal motion relative thereto; a plunger piston operatively coupled to the ball nut; a block housing at least partially enclosing the plunger piston and the ball screw; a plunger chamber at least partially defined by the block housing and a face of the plunger piston, the plunger chamber being configured to contain hydraulic fluid and selectively pressurized by reciprocal motion of the plunger piston with respect to the block housing driven by longitudinal motion of the ball nut; an output channel selectively placing the plunger chamber in fluid communication with at least one component of a brake system including the power transmission unit; an electric motor for selectively driving the ball screw to responsively reciprocate the plunger piston within the plunger chamber; and a motor housing at least partially enclosing the ball nut, the electric motor, and the

plunger piston, the motor housing including a block lip which is crimped to the block housing to maintain the motor housing in relation to the block housing; the plunger piston defines defining a longitudinal internal plunger cavity for receiving at least a portion of the ball screw, the internal plunger cavity including a force dissipator at a blind end thereof, the force dissipator being configured for selective contact with a cantilevered end of the ball screw.

16. The power transmission unit of claim 15, wherein the force dissipator includes a spring member.

17. The power transmission unit of claim 15, wherein the force dissipator includes a resilient bumper.

18. The power transmission unit of claim 17, wherein the force dissipator includes a planar member interposed longitudinally between the resilient bumper and the cantilevered end of the ball screw.

19. The power transmission unit of claim 15, wherein the cantilevered end of the ball screw includes a convex profile configured to concentrate contact with the force dissipator at a predetermined area of the convex profile.

20. The power transmission unit of claim 15, wherein the plunger chamber defines a longitudinally extending central chamber axis which is laterally offset from a longitudinally extending central plunger axis of the plunger piston.

21. The power transmission unit of claim 15, wherein the electric motor includes a motor rotor operatively coupled to the ball screw and a motor stator mounted on the motor housing wherein torque developed by the motor stator is resisted by at least a portion of the motor housing.

22. A hydraulic brake by wire unit, comprising the power transmission unit of claim 15; first and second multiplex control valve arrangements fluidly connected to the output channel of the power transmission unit, each of the first and second multiplex control valve arrangements being interposed hydraulically between the power transmission unit and a respective one of first and second iso tube ports, the first and second multiplex control valve arrangements each including respective first and second iso-type valves.
